The old basket model, where it fails, and the mixture model that replaces it

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Part 1 states the old model and how it is used. Part 2 locates its failure. Part 3 builds the mixture model step by step and compares the two.

1 The old model

1.1 What is modelled

One shopping trip is fixed throughout. $\mathcal{J}$ is the set of products offered on that trip, and each product $j \in \mathcal{J}$ belongs to one category $c(j)$.

A *basket* is the set $S \subseteq \mathcal{J}$ of products bought on the trip. For a basket we write $n(S) = |S|$ for its size, $n_c(S)$ for the number of its products in category c, and $m(S)$ for the number of categories with $n_c(S) > 0$. The allowed baskets form the set

$$\mathcal{S} = \{S \subseteq \mathcal{J} : \ 1 \le n(S) \le 120, \ n_c(S) \le 40 \text{ for every } c\}.$$

A model is a probability $p(S)$ for every $S \in \mathcal{S}$, with $\sum_{S \in \mathcal{S}} p(S) = 1$.

1.2 The parameters

- $b_j \in \mathbb{R}$: the utility of product j on this trip, built from its popularity, price, promotions, season, store and the household's tastes and history. On a given trip it is a fixed number.
- $\phi_j \in \mathbb{R}^r$: the *loading* of product j, a vector of dimension r ($r = 16$ on the real data). Products with similar loadings tend to be bought together.
- $\psi_c \in \mathbb{R}^r$: a loading for each product of category c beyond the first.
- ρ_c, $f_c(k)$, $\rho_0(m)$: penalties on pairs within category c, on holding k products of c, and on having m categories in the basket. $(a)_+ = \max(a, 0)$ and $\binom{k}{2} = k(k-1)/2$.

1.3 The definition

Each basket gets two numbers. The first is the sum of its product utilities and penalties,

$$a(S) = \sum_{j \in S} \big(b_j - \tfrac{1}{2}\|\phi_j\|^2\big) - \sum_c \Big[\rho_c\big(\tbinom{n_c(S)}{2}\big) + f_c\big(n_c(S)\big) - \tfrac{1}{2}(n_c(S)-1)_+\|\psi_c\|^2\Big] - \rho_0\big(m(S)\big).$$

The second is the basket's direction in loading space,

$$s(S) = \sum_{j \in S} \phi_j - \sum_c (n_c(S) - 1)_+ \, \psi_c \ \in \mathbb{R}^r.$$

Definition 1 (old model)**.**

$$p_{\text{old}}(S) = \frac{e^{F(S)}}{Z_{\text{old}}}, \qquad F(S) = a(S) + \tfrac{1}{2}\|s(S)\|^2, \qquad Z_{\text{old}} = \sum_{S \in \mathcal{S}} e^{F(S)}. \tag{1}$$

$e^{F(S)}$ is the *score* of basket S. Z_{old}, the *normalising constant*, is the sum of the scores of all allowed baskets; dividing by it makes the probabilities add up to one. A basket's probability is its share of the total score.

The term $\frac{1}{2}\|s(S)\|^2$ is what makes products go together. Ignoring ψ, it equals $\frac{1}{2}\sum_{j\in S}\|\phi_j\|^2 + \sum_{\{i,j\}\subseteq S}\phi_i\cdot\phi_j$: every pair of products in the basket adds the dot product of their loadings, so products with similar loadings raise each other's probability. The old model has no prior and no hidden variable: (1) scores baskets directly.

1.4 How the model is used

Training maximises the log-likelihood $\sum_t \log p_{\text{old}}(S_t)$ over the observed baskets S_t of the training trips. For any parameter ϑ,

$$\nabla_\vartheta \log p_{\text{old}}(S) = \nabla_\vartheta F(S) - \mathbb{E}_{S'\sim p_{\text{old}}}\big[\nabla_\vartheta F(S')\big], \tag{2}$$

"observed minus what the model expects". *Scoring* a held-out basket needs $\log Z_{\text{old}}$, and *generating* baskets needs draws from p_{old}. All three need sums over the whole of $\mathcal{S}$, which is far too large to list.

The taste variable. The pair term couples every product with every other, so these sums cannot be done directly. They become tractable after one identity.

Lemma 1 (Gaussian identity)**.** *Let $\mathcal{N}(z;0,I) = (2\pi)^{-r/2}e^{-\|z\|^2/2}$ be the standard Gaussian density on $\mathbb{R}^r$. For every $s\in\mathbb{R}^r$,*

$$e^{\frac{1}{2}\|s\|^2} = \int_{\mathbb{R}^r} \mathcal{N}(z;0,I)\, e^{z\cdot s}\, dz.$$

Proof. $-\frac{1}{2}\|z\|^2 + z\cdot s = -\frac{1}{2}\|z-s\|^2 + \frac{1}{2}\|s\|^2$, and $\mathcal{N}(z;s,I)$ integrates to one. □

Apply it to the score: $e^{F(S)} = \int \mathcal{N}(z;0,I)\, e^{a(S)+z\cdot s(S)}\, dz$. Inside the integral the pair term has gone; for a fixed vector z, called the *taste*, the exponent $a(S)+z\cdot s(S)$ is a sum over products and categories. This suggests looking at baskets one taste at a time.

Definition 2 (basket law given a taste)**.** *For $z\in\mathbb{R}^r$,*

$$p(S\mid z) = \frac{e^{E_z(S)}}{Z(z)}, \qquad E_z(S) = a(S) + z\cdot s(S), \qquad Z(z) = \sum_{S\in\mathcal{S}} e^{E_z(S)}. \tag{3}$$

$Z(z)$ is the normalising constant of this law: the total score of all allowed baskets when the taste is z. Writing out $E_z(S)$, each product contributes $b_j - \frac{1}{2}\|\phi_j\|^2 + z\cdot\phi_j$: the taste raises the products whose loadings point its way. Because E_z has no pair term, $Z(z)$ and draws from $p(S\mid z)$ are computed *exactly* by a dynamic program (category by category, tracking the basket size and the number of categories).

Proposition 1 (the old model through tastes)**.**

$$p_{\text{old}}(S) = \int_{\mathbb{R}^r} \pi_{\text{old}}(z)\, p(S\mid z)\, dz, \qquad \pi_{\text{old}}(z) = \frac{\mathcal{N}(z;0,I)\, Z(z)}{Z_{\text{old}}}. \tag{4}$$

π_{old} is a probability density on $\mathbb{R}^r$.

Proof. $e^{F(S)} = \int \mathcal{N}(z;0,I)\, e^{E_z(S)} dz = \int \mathcal{N}(z;0,I)\, Z(z)\, p(S\mid z)\, dz$; divide by Z_{old}. Summing over S, $\int \pi_{\text{old}}(z)\, dz = \sum_S p_{\text{old}}(S) = 1$. □

So every computation with the old model is an average over tastes z drawn from π_{old}, with an exact inner part for each z. In particular the expectation in (2) is $\int \pi_{\text{old}}(z)\, \mathbb{E}_{S'\sim p(\cdot\mid z)}[\cdots]\, dz$. In practice the average over z was taken with Gibbs chains that alternate $z\sim\mathcal{N}(s(S),I)$ and $S\sim p(\cdot\mid z)$, started from the observed baskets.

2 Where exactly the problem is

The problem is the taste density $\pi_{\text{old}}(z) \propto \mathcal{N}(z; 0, I)\, Z(z)$: the Gaussian density multiplied by the total score $Z(z)$. This section shows that $Z(z)$ grows fastest for tastes that make large baskets attractive, that this can give π_{old} several separate peaks, and that the procedures above see only one of them.

2.1 $Z(z)$ grows fastest for large baskets

Example 1. Three products A, B, C, one taste dimension ($r = 1$), loadings $\phi_A = \phi_B = \phi_C = 1$, utilities $b_A = 1.5$, $b_B = 0.5$, $b_C = -0.5$, no penalties. A product's contribution is $b_j - \frac{1}{2} + z$, so at $z = 0$ the three contributions are $1, 0, -1$ and at $z = 2$ they are $3, 2, 1$. The scores $e^{E_z(S)}$:

S	$\{A\}$	$\{B\}$	$\{C\}$	$\{A,B\}$	$\{A,C\}$	$\{B,C\}$	$\{A,B,C\}$	$Z(z)$
$z = 0$	2.72	1.00	0.37	2.72	1.00	0.37	1.00	9.17
$z = 2$	20.1	7.4	2.7	148.4	54.6	20.1	403.4	656.7

Moving the taste from 0 to 2 multiplies the score of every basket of n products by e^{2n}. Z grows 72-fold, mostly through the largest basket, whose probability $p(\{A, B, C\} \mid z)$ rises from $1.00/9.17 = 0.11$ to $403.4/656.7 = 0.61$.

In general. Write $\mathbb{E}[s \mid z] = \sum_S p(S \mid z)\, s(S)$ and $\text{Cov}[s \mid z] = \sum_S p(S \mid z)\, (s(S) - \mathbb{E}[s \mid z])(s(S) - \mathbb{E}[s \mid z])^\top$ for the mean and covariance of the basket's direction under $p(\cdot \mid z)$.

Lemma 2. $\nabla_z \log Z(z) = \mathbb{E}[s \mid z]$ *and* $\nabla_z^2 \log Z(z) = \text{Cov}[s \mid z]$.

Proof. $\log Z(z) = \log \sum_S e^{a(S) + z \cdot s(S)}$; differentiate once and twice. □

Moving z in a direction raises $\log Z$ at the rate of the expected basket direction along it, and that rate itself grows as z moves on. With many products whose loadings point the same way, $\mathbb{E}[s \mid z]$ keeps growing until the baskets hold all of them.

2.2 π_{old} can have several peaks

Proposition 2.

$$\nabla_z \log \pi_{\text{old}}(z) = -z + \mathbb{E}[s \mid z], \qquad \nabla_z^2 \log \pi_{\text{old}}(z) = -I + \text{Cov}[s \mid z].$$

Proof. $\log \pi_{\text{old}}(z) = -\frac{1}{2}\|z\|^2 + \log Z(z) + \text{const}$; apply Lemma 2. □

Two forces set the slope: $-z$ from the Gaussian pulls the taste back towards zero, and $\mathbb{E}[s \mid z]$ from $Z(z)$ pushes it outward, harder the further out it is. Wherever $\text{Cov}[s \mid z]$ exceeds I the curvature is positive, so $\log \pi_{\text{old}}$ need not be concave and can have several local maxima (peaks). Every peak solves $z = \mathbb{E}[s \mid z]$.

Example 2. Twenty products in twenty categories, $r = 1$, all with $b_j = -4$ and $\phi_j = 1$, no penalties. Each product contributes $-4.5 + z$, the basket's direction is its size $n(S)$, and

$$Z(z) = \left(1 + e^{-4.5+z}\right)^{20} - 1,$$

the -1 removing the empty basket. The zeros of the slope $-z + \mathbb{E}[n \mid z]$, computed numerically:

z	$\mathbb{E}[n \mid z]$	$\log \pi_{\text{old}}(z) + \text{const}$	
1.56	1.56	-0.63	peak 1: one or two products per basket
2.36	2.36	-0.68	valley
20.0	20.0	$+110.0$	peak 2: all 20 products

Peak 2 is $e^{110.6}$ times higher. Integrating (4) numerically, tastes $z < 5$ carry a fraction 10^{-46} of the probability: under p_{old} essentially every basket holds all 20 products. The same is visible in (1) directly: the full basket has $a = 20 \times (-4.5) = -90$ and pair bonus $\frac{1}{2} \cdot 20^2 = 200$, so $F = 110$, against $F = -4$ for a single product. The pair bonus grows like n^2, the penalties only like n.

2.3 Why training and scoring did not see it

Suppose the real baskets hold one or two products. The Gibbs chains start at the observed baskets, so their tastes start near peak 1. To reach peak 2 a chain must cross the valley, which is improbable at every step, and the chains ran a few sweeps. So:

- the expectation in (2) is computed over peak 1 only; nothing in training sees, or lowers, the probability at peak 2;
- the estimate of Z_{old}, built from the same chains, leaves out peak 2, so $\log p_{\text{old}}$ of held-out baskets is reported too high.

On the Dunnhumby refit peak 2 was the region of 120-product baskets: the reported held-out log-likelihood was -38.8 per trip and -53.8 once the far peak was included.

The fault is therefore not in $p(S \mid z)$, which is exact, but in the taste density $\pi_{\text{old}} \propto \mathcal{N}(z; 0, I)\, Z(z)$: a taste gains weight in proportion to the total score of the baskets it allows.

3 The mixture model

The mixture model keeps the exact basket law given a taste and changes only how tastes are weighted.

3.1 What is modelled

As before, the basket $S \in \mathcal{S}$ bought on one trip. In addition, every trip has an unobserved taste $z \in \mathbb{R}^r$: what the shopper is in the mood for on that trip. The model describes how z arises and how S arises given z.

3.2 The prior on the taste

$$z \sim \mathcal{N}(0, I), \qquad \text{independently on every trip.} \tag{5}$$

Most tastes are mild (near zero); strong ones are rare. Lasting preferences of a household are not in z but in the utilities b_j.

3.3 The basket given the taste

Given z, the basket follows the law (3) of Part 1:

$$p(S \mid z) = \frac{e^{E_z(S)}}{Z(z)}, \qquad E_z(S) = a(S) + z \cdot s(S),$$

where the normalising constant

$$Z(z) = \sum_{S \in \mathcal{S}} e^{E_z(S)}$$

is the total score of all allowed baskets under taste z, so that $\sum_S p(S \mid z) = 1$ for every z. $Z(z)$ and draws from $p(\cdot \mid z)$ are computed exactly by the dynamic program.

3.4 The probability of a basket

Averaging the basket law over the prior:

$$p(S) = \int_{\mathbb{R}^r} \mathcal{N}(z; 0, I)\, p(S \mid z)\, dz. \tag{6}$$

It sums to one because each $p(\cdot \mid z)$ does. To generate a basket, draw z from (5) and then S from $p(\cdot \mid z)$; both steps are exact, with no Markov chain.

Example 2 under the mixture model. The taste $z = 20$ has prior density proportional to e^{-200} and is never drawn. Typical tastes lie in $[-2, 2]$, where each product is bought with probability between 0.0015 and 0.076. The mean basket size is $\int \mathcal{N}(z; 0, 1)\, \mathbb{E}[n \mid z]\, dz = 1.19$, and the full 20-product basket has probability 2.3×10^{-10}.

3.5 The posterior of the taste

Scoring an observed basket S, and training on it, needs the tastes that could have produced it. By Bayes' rule their density is

$$p(z \mid S) = \frac{\mathcal{N}(z; 0, I)\, p(S \mid z)}{p(S)}, \qquad \log p(z \mid S) = -\tfrac{1}{2}\|z\|^2 + z \cdot s(S) - \log Z(z) + \text{const}, \tag{7}$$

where the constant collects $a(S)$ and $\log p(S)$ and does not depend on z.

Proposition 3 (one peak)**.**

$$\nabla_z \log p(z \mid S) = -z + s(S) - \mathbb{E}[s \mid z], \qquad \nabla_z^2 \log p(z \mid S) = -I - \operatorname{Cov}[s \mid z] \preceq -I.$$

Hence $\log p(z \mid S)$ *is strictly concave and has exactly one maximum* $\hat{z}$*; moreover* $\log p(z \mid S) \leq \log p(\hat{z} \mid S) - \frac{1}{2}\|z - \hat{z}\|^2$ *for every* z*.*

Proof. Differentiate (7) using Lemma 2. A covariance is positive semi-definite, so the Hessian is at most $-I$ in every direction; a function with this property has exactly one maximum and the stated quadratic bound. □

Compare with Proposition 2. The term $\mathbb{E}[s \mid z]$ that pushed π_{old} outward enters here with a minus sign, so it pulls in the same direction as the prior instead of against it. The observed basket contributes only the fixed vector $s(S)$, which moves the peak but cannot create a second one. This holds for every basket, every value of the parameters and every dimension r. In Example 2, for a basket of one product the slope is $-z + 1 - \mathbb{E}[n \mid z]$; both terms decrease in z, and the single peak is at $\hat{z} = -0.10$.

Consequences.

- *Scoring:* $p(S)$ is estimated by importance sampling around $\hat{z}$, which gradient ascent always finds.
- *Training:* the gradient is a weighted average, over tastes drawn around $\hat{z}$, of "observed minus expected given that taste". The expectation given a taste is exact, so no region of baskets is left unvisited.
- *Generation* is exact, as above.

3.6 The two models side by side

The old model implies the taste density $\pi_{\text{old}}(z) = \mathcal{N}(z; 0, I)\, Z(z)/Z_{\text{old}}$ (Proposition 1). It is not a prior chosen in advance: it depends on all the parameters and on the trip, and it favours tastes with a large total score. The mixture model uses the prior $\mathcal{N}(0, I)$ itself.

	old model	mixture model
basket law given a taste	$p(S \mid z)$, exact	the same
weight of taste z	$\mathcal{N}(z; 0, I)\, Z(z)/Z_{\text{old}}$	$\mathcal{N}(z; 0, I)$
density searched when scoring	$\pi_{\text{old}}(z)$: several peaks possible	$p(z \mid S)$: exactly one peak
generating a basket	Markov chain	exact: z, then $S \mid z$

What changes, and what does not.

- The two are different models with the same parameters; a fit of one is not a fit of the other.
- In the old model a price change alters $Z(z)$ and so shifts which tastes are common. In the mixture model the distribution of tastes is fixed by the prior, and prices change only what each taste buys.
- The mixture model removes the hidden peaks; it does not by itself make training correct. In the full real fit the held-out log-likelihood on all 1,024 validation trips fell from -38.10 at step 240 to -38.76 at the end (paired difference -0.66 ± 0.08). That is a training problem, to be analysed separately.